

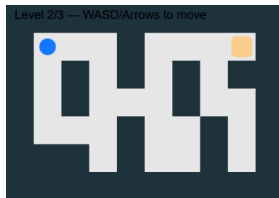
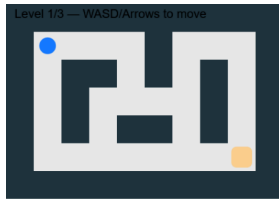
Process & Decision Documentation

Project/Assignment Decisions

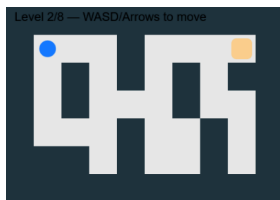
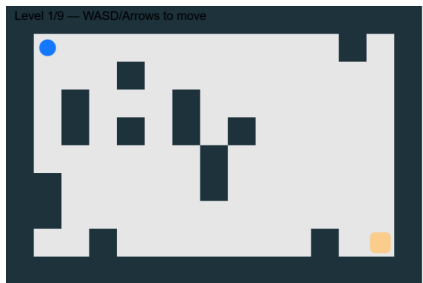
Side Quests and A4 (Individual Work)

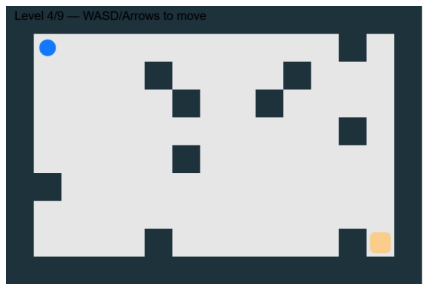
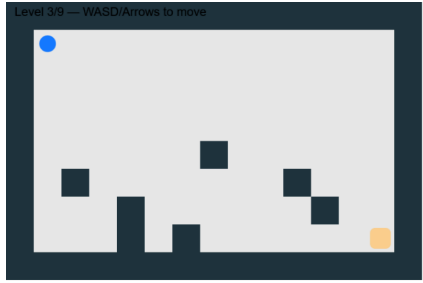
Added a feature to generate levels using array data and loops to dynamically place tiles. The number of levels is randomly generated and consists of a larger 10 by 15 grid. I changed the level dynamics to include a wider open space for speedrun times.

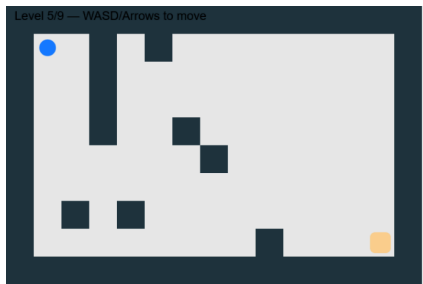
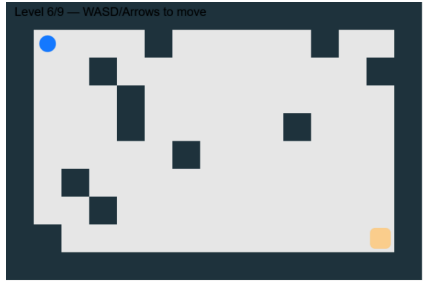
Default game (States):

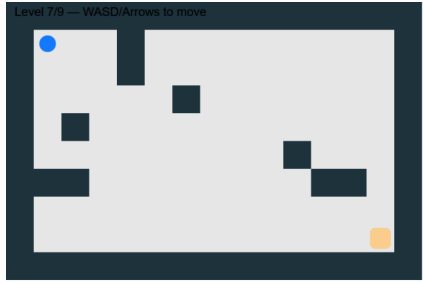


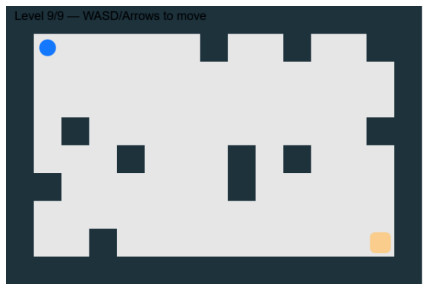
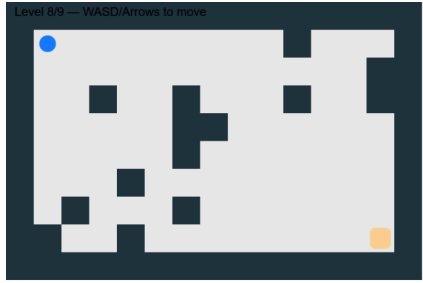
Modified game:











Code Changes

Created function

```
✓ 2 sketch.js ...  
... @@ -131,3 +131,5 @@ function copyGrid(grid) {  
131 131 */  
132 132 return grid.map((row) => row.slice());  
133 133 }  
134 +  
135 + function generateLevel(rows, cols) {}
```

Filled out function

```
✓ 26 sketch.js ...  
... @@ -132,4 +132,28 @@ function copyGrid(grid) {  
132 132 return grid.map((row) => row.slice());  
133 133 }  
134 134  
135 - function generateLevel(rows, cols) {}  
135 + function generateLevel(rows, cols) {  
136 + let newGrid = [];  
137 +  
138 + for (let r = 0; r < rows; r++) {  
139 + let row = [];  
140 +  
141 + for (let c = 0; c < cols; c++) {  
142 + let tile = 0;  
143 +  
144 + if (r === 0 || r === rows - 1 || c === 0 || c === cols - 1) {  
145 + tile = 1;  
146 + } else if (Math.random() < 0.1 && (r > 2 || c > 2)) {  
147 + tile = 1;  
148 + }  
149 +  
150 + row.push(tile);  
151 + }  
152 + newGrid.push(row);  
153 + }  
154 +  
155 + newGrid[1][1] = 2;  
156 + newGrid[rows - 2][cols - 2] = 3;  
157 +  
158 + return newGrid;  
159 + }
```

New variable

```
✓ 2 sketch.js ...  
... @@ -43,6 +43,8 @@ function setup() {  
43 43 */  
44 44 levels = levelsData.levels.map((grid) => new Level(copyGrid(grid), TS));  
45 45  
46 + let dynamicGrid = generateLevel(10, 15);  
47 +  
46 48 // Create a player.  
47 49 player = new Player(TS);  
48 50  
...  
↓
```

Added command line

```
sketch.js

@@ -45,6 +45,8 @@ function setup() {
45 45
46 46     let dynamicGrid = generateLevel(10, 15);
47 47
48 48 +   levels.push(new Level(dynamicGrid, TS));
49 49 +
48 50     // Create a player.
49 51     player = new Player(TS);
50 52

```

Random calls for maze dimensions

```
sketch.js

@@ -43,9 +43,13 @@ function setup() {
43 43     */
44 44     levels = levelsData.levels.map((grid) => new Level(copyGrid(grid), TS));
45 45
46 46 -   let dynamicGrid = generateLevel(10, 15);
46 46 +   let randomGrid = generateLevel(Math.random(10, 20), Math.random(15, 30));
47 47 +   levels[0] = new Level(randomGrid, TS);
47 48
48 48 -   levels.push(new Level(dynamicGrid, TS));
49 49 +   for (let levelCount = 0; levelCount < Math.random(1, 5); levelCount++) {
50 50 +       randomGrid = generateLevel(Math.random(10, 20), Math.random(15, 30));
51 51 +       levels.push(new Level(randomGrid, TS));
52 52 +   }
49 53
50 54     // Create a player.
51 55     player = new Player(TS);

```

Revert to random 3 range

```
sketch.js

@@ -43,11 +43,11 @@ function setup() {
43 43     */
44 44     levels = levelsData.levels.map((grid) => new Level(copyGrid(grid), TS));
45 45
46 46 -   let randomGrid = generateLevel(Math.random(10, 20), Math.random(15, 30));
46 46 +   let randomGrid = generateLevel(10, 15);
47 47     levels[0] = new Level(randomGrid, TS);
48 48
49 49 -   for (let levelCount = 0; levelCount < Math.random(1, 5); levelCount++) {
50 50 -       randomGrid = generateLevel(Math.random(10, 20), Math.random(15, 30));
49 49 +   for (let levelCount = 0; levelCount < Math.random(3); levelCount++) {
50 50 +       randomGrid = generateLevel(10, 15);
51 51         levels.push(new Level(randomGrid, TS));
52 52     }
53 53

```

Random level number generation



```
sketch.js

function setup() {
  let randomGrid = generateLevel(10, 15);
  levels[0] = new Level(randomGrid, TS);
  for (let levelCount = 0; levelCount < Math.random(3); levelCount++) {
    let randomCount = int(random(3, 9));
    for (let levelCount = 0; levelCount < randomCount; levelCount++) {
      randomGrid = generateLevel(10, 15);
      levels.push(new Level(randomGrid, TS));
    }
  }
}
```

GenAI Documentation

No GenAI was used.